Computational Learning Theory

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Contents

- Maximum Likelihood
- Maximum A Posteriori，MAP
- PAC learning
  - Sample complexity
  - Computational complexity

Maximum Likelihood

Supposed that the data $D = \{x_i\}_{i=1}^N$ is sampled from unknown probability $p(x \mid \theta)$.
The task is to estimate the parameter θ from training data.

General learning principle: **Maximum Likelihood**

- Given training data $D = \{x_i\}_{i=1}^N$, estimate the parameter θ by maximizing its likelihood

$$\hat{\theta} = \arg\max_{\theta} \prod_{i=1}^{N} p(x_i \mid \theta)$$

where $L(\theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$ is its likelihood.

Maximum Likelihood

- Maximizes $L(\theta)$ by solving the following equation

$$\left.\frac{\partial \ln L(\theta)}{\partial \theta}\right|_{\theta=\theta^*} = 0$$

  – Often easier to maximize $\ln L(\theta)$

  – $L(\theta)$ and $\ln L(\theta)$ achieve their maximum **at the same location**.

- $\ln L(\theta)$ converts the product into a summation.

$$\ln L(\theta) = \sum_{i=1}^{N} \ln p(x_i, \theta)$$

Log Maximum Likelihood

- The new maximization condition is:

$$\left.\frac{\partial \ln L}{\partial \theta}\right|_{\theta=\theta^*} = \sum_{i=1}^{N} \frac{\partial}{\partial \theta} \ln p(x_i, \theta)\Big|_{\theta=\theta^*} = 0$$

- **Resultant equations**: simple linear equations or coupled non-linear equations.

- **Example:** Let $p(x,\theta)$ be given by a Gaussian distribution function.
- Let $\theta = \mu$ be the mean of the Gaussian.

- To find the best estimate of θ from our set of n easurements .

$$\{x_1, x_2, \ \ldots, x_N\}$$

Example 1: Gaussian

- Gaussian PDF

$$f(x,\alpha)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(x-\alpha)^2}{2\sigma^2}\right)$$

- The likelihood function for this problem is:

$$L=\prod_{i=1}^{n} f(x_i,\alpha)=\prod_{i=1}^{n}\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\sum_{i=1}^{n}\frac{(x_i-\alpha)^2}{2\sigma^2}\right)$$

$$=\left[\frac{1}{\sigma\sqrt{2\pi}}\right]^n e^{-\frac{(x_1-\alpha)^2}{2\sigma^2}} e^{-\frac{(x_2-\alpha)^2}{2\sigma^2}}\cdots e^{-\frac{(x_n-\alpha)^2}{2\sigma^2}}=\left[\frac{1}{\sigma\sqrt{2\pi}}\right]^n e^{-\sum_{i=1}^{n}\frac{(x_i-\alpha)^2}{2\sigma^2}}$$

Example 1: Gaussian

$$\ln L = \ln \prod_{i=1}^{n} f(x_i, \alpha) = \ln\left([\frac{1}{\sigma\sqrt{2\pi}}]^n \exp\left(-\sum_{i=1}^{n} \frac{(x_i - \alpha)^2}{2\sigma^2} \right) \right)$$

$$= n \ln(\frac{1}{\sigma\sqrt{2\pi}}) - \sum_{i=1}^{n} \frac{(x_i - \alpha)^2}{2\sigma^2}$$

- We want to find the *a* that maximizes the log likelihood function:

$$\frac{\partial \ln L}{\partial \alpha} = \frac{\partial}{\partial \alpha}\left[n \ln\left(\frac{1}{\sigma\sqrt{2\pi}} \right) - \sum_{i=1}^{n} \frac{(x_i - \alpha)^2}{2\sigma^2} \right] = 0$$

$$\frac{\partial}{\partial \alpha} \sum_{i=1}^{n} (x_i - \alpha)^2 = 0; \qquad \sum_{i=1}^{n} 2(x_i - \alpha)(-1) = 0, \quad \alpha = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Example 2: Regression formulation

- Consider a linear expansion

$$y = f_\beta(x) = \sum_{j=1}^{L} \beta_j h_j(x) + \varepsilon, \qquad f(\varepsilon, \sigma^2) = \frac{1}{\sqrt{2\pi}\ \sigma} e^{-\frac{\varepsilon^2}{2\sigma^2}}$$

- Conditional probability:

$$p(y \mid x, \beta) = N(f_\beta(x), \sigma^2)$$

- The log-likelihood

$$l(\theta) = -\frac{N}{2}\log \sigma^2 2\pi - \boxed{\frac{1}{2\sigma^2}\sum_{i=1}^{N}\left(y_i - \beta^T h(x_i)\right)^2}$$

- Estimating equations:

$$\frac{\partial l(\theta)}{\partial \beta} = 0; \qquad \frac{\partial l(\theta)}{\partial \sigma^2} = 0.$$

Regression

- Consider a linear expansion: $\mu(x) = \sum_{j=1}^{N} \beta_j h_j(x)$

- The least square error solution: $\hat{\beta} = (H^T H)^{-1} H^T y$

- The Covariance of $\hat{\beta}$

$$cor(\hat{\beta}) = (H^T H)^{-1} \hat{\sigma}^2; \qquad \hat{\sigma}^2 = \sum (y_i - \hat{\mu}(x_i))^2 / N$$

General properties of MLM

- For large data samples (large *n*) the likelihood function, *L*, approaches a Gaussian distribution.
- Maximum likelihood estimates are usually *consistent*.
  - For large *n* the estimates converge to the true value of the parameters we wish to determine.
- Maximum likelihood estimates are usually *unbiased*.
  - For all sample sizes the parameter of interest is calculated correctly.
- Maximum likelihood estimate is *efficient*: the estimate has the smallest variance.
- Maximum likelihood estimate is *sufficient*: it uses all the information in the observations (the x_i's).

- Bad news: we must know the correct probability distribution for the problem at hand!

Maximum Likelihood and Posteriori Maximazation

- Given training data $D = \{x_i\}_{i=1}^{N}$, estimate parameter θ by maximizing its likelihood

$$\hat{\theta} = \arg\max_{\theta} \prod_{i=1}^{N} p(x_i \mid \theta)$$

where $p(x \mid \theta)$ is its probabilty density function

- Assume that $p(\theta)$ is the prior distribution for θ. The better approach for estimating the parameter θ is Posteriori Maximazation

$$\hat{\theta} = \arg\max_{\theta} p(\theta \mid D) = \arg\max_{\theta} p(D \mid \theta) p(\theta)$$

Likelihood prior

Regularization as Posteriori Maximazation

- Consider the regression problem for a parametric model

$$y = f(x \mid \theta) + \varepsilon, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2)$$

Given training data $D = \{(x_i, y_i)\}_{i=1}^N$, find the parameter θ by maximizing its likelihood

$$\hat{\theta} = \arg\max_\theta \prod_{i=1}^N p(y_i \mid x_i, \theta)$$

- If the error term follows $\varepsilon \sim N(0, \sigma_\varepsilon^2)$, then the maximum likelihood is equivalent to the RSS minimization.

- Posteriori Maximazation:

$$\hat{\theta} = \arg\max_\theta p(\theta \mid D) = \arg\max_\theta \prod_{i=1}^N p(y_i \mid x_i, \theta) p(\theta)$$

$$\boxed{= \arg\max_\theta \sum_{i=1}^N \log p(y_i \mid x_i, \theta) + \log p(\theta)}$$

An Example

- Consider a linear expansion

$$y = \mu_\beta(x) + \varepsilon = \sum_{j=1}^{N} \beta_j h_j(x) + \varepsilon; \qquad p(\varepsilon) = \frac{1}{(2\pi)^{1/2}\sigma_\varepsilon} \exp(-\tfrac{\varepsilon^2}{2\sigma^2})$$

- The prior probability $\beta \propto N(0, \tau \sum)$

- The maximum a posterior

$$p(\beta | y, x) \approx p(\beta)\, p(y \mid x, \beta) = N(0, \tau \sum) N_y(\mu_\beta(x) \mid \sigma_\varepsilon^2)$$

MAP (Maximum A Posterior)

- The posterior distribution for β is also Gaussian, with mean and covariance

$$\mathrm{E}(\beta \mid \mathbf{Z}) = \left(\mathbf{H}^T\mathbf{H} + \frac{\sigma^2}{\tau}\Sigma^{-1}\right)^{-1}\mathbf{H}^T\mathrm{y}, \qquad \mathrm{cov}(\beta \mid \mathbf{Z}) = \left(\mathbf{H}^T\mathbf{H} + \frac{\sigma^2}{\tau}\Sigma^{-1}\right)^{-1}\sigma^2.$$

- The corresponding posterior values for $\mu(x) = \sum_{j=1}^{N} \beta_j h_j(x)$

$$\mathrm{E}(\mu(x) \mid \mathbf{Z}) = h(x)^T\left(\mathbf{H}^T\mathbf{H} + \frac{\sigma^2}{\tau}\Sigma^{-1}\right)^{-1}\mathbf{H}^T\mathrm{Y},$$

$$\mathrm{cov}\left[\mu(x), \mu(x') \mid \mathbf{Z}\right] = h(x)^T\left(\mathbf{H}^T\mathbf{H} + \frac{\sigma^2}{\tau}\Sigma^{-1}\right)^{-1}h(x')\sigma^2.$$

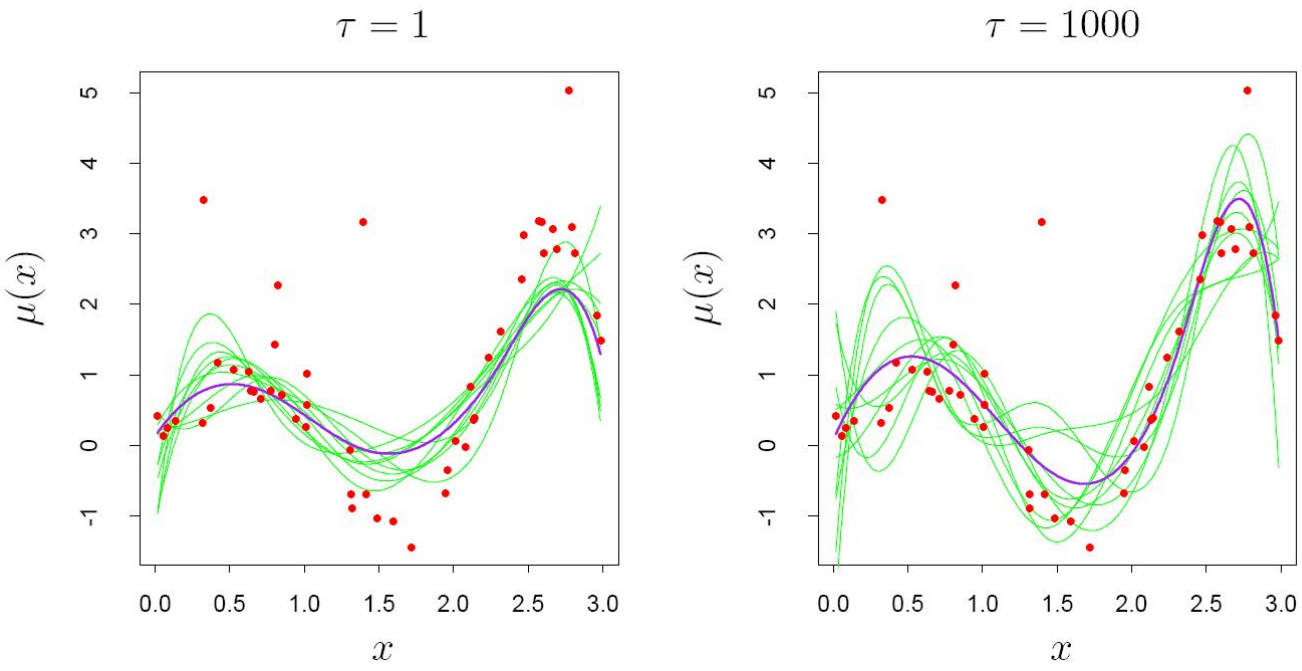

FIGURE 8.4. *Smoothing example: Ten draws from the posterior distribution for the function $\mu(x)$, for two different values of the prior variance τ. The purple curves are the posterior means.*

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  - Esitmation of Generalization Error

Generalizability of Learner

- In machine learning, we care about the generalization error, but most learning algorithms fit their models to the training set.

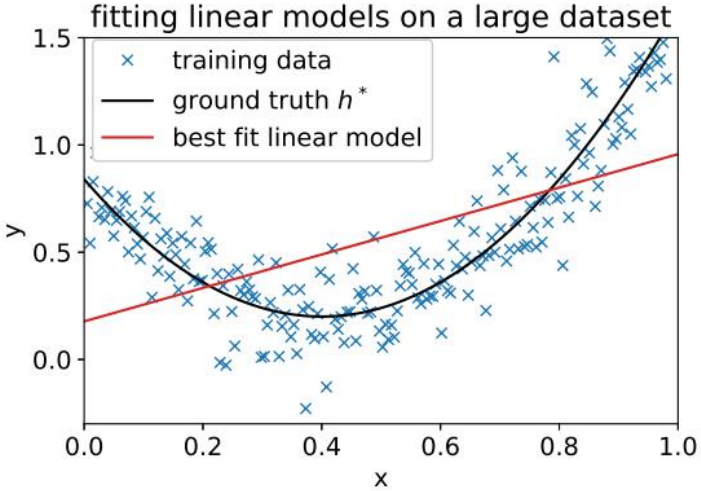
- Relation between the training error and the generalization error?
- Are there conditions under which we can actually prove that learning algorithms will work well?

What General Laws for Inductive Learning?

- **Sample complexity**
  - How many training examples are sufficient to learn the target concept?
- **Computational complexity**
  - Resources required to learn the target concept?
- **What theory to relate:**
  - Training examples (number: m)
    - Quantity
    - Quality
    - How to present
  - Complexity of hypothesis / concept space: H
  - Accuracy of approx to target concept: ε
  - Probability of successful learning: $1-\delta$

Two Basic Models

PAC Framework	Agnostic Framework
Sample labels are consistent with some hypothesis h in H	No prior restriction on the sample labels
Learner's hypothesis are required to meet abslute upper bound on its error	The required upper bound on its hypothesis error is only relative (to the best hypothesis in the class)



Probably Approximately Correct

- **Goal:**
  - PAC-Learner produces hypothesis $\hat{h}$ that is approximately correct

$$err_D(\hat{h}) \approx 0$$

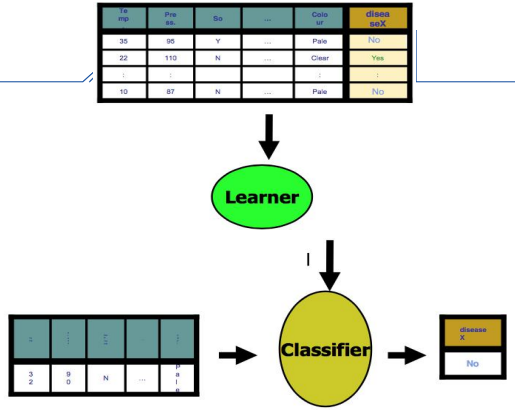
with high probability

$$p\left(err_D(\hat{h}) \approx 0\right) \approx 1$$

- **Double "hedging"**
  - approximately (error)
  - probably

Need both !

Formulation



- **Given:**
  - set of examples $\mathbf{X}$
  - fixed (unknown) distribution $\mathcal{D}$ over $\mathbf{X}$
  - set of hypotheses $\mathbf{H}$
  - set of possible target concepts C

- **Learner observes sample S =** $\left\{\left(x^{(i)}, c^{(i)}\right)\right\}_{i=1}^{N}$
  - Instances $x^{(i)}$ drawn from distribution $\mathcal{D}$
  - labeled by target concept $\mathrm{c} \in \mathrm{C}$

- **Learner outputs** $h \in H$ estimating c
  - h is evaluated by performance on subsequent instances drawn from $\mathcal{D}$
  - If $C = H$ (so $\mathrm{c} \in \mathrm{H}$); noise-free data

Two Notions of Error

- **Training error** (a.k.a., empirical risk or empirical error) of hypothesis h with respect to target concept c
  - How often $h(x) \neq c(x)$ over training instance from S

$$\hat{\varepsilon}_S(h) \equiv p_{x \in S}\left[c(x) \neq h(x)\right] \equiv \sum_{x \in S} \delta\left[c(x) \neq h(x)\right] \Big/ |S|$$

- **True error of** (a.k.a., generalization error, test error) hypothesis h with respect to c

  - How often $h(x) \neq c(x)$ over future random instances drew i.i.d. from D

$$\varepsilon_D(h) \equiv p_{x \in D}\left[c(x) \neq h(x)\right]$$

Can we bound

$\hat{\epsilon}_{\mathcal{D}}(h)$

in terms of

$\hat{\epsilon}_{\mathcal{S}}(h)$

??

PAC learnable

- Definition (PAC Learnability)
  - A hypothesis H is known as PAC Learnable, if for any $\varepsilon, \delta \in (0,1)$, there exists a function $N(\varepsilon, \delta)$

$$N(\varepsilon, \delta): (0,1)^2 \to N$$

for any integer $N \geq N(\varepsilon, \delta)$, the learner can find a predictor $\hat{h} \in H$, learned from sample data, such that

$$p(\varepsilon_D(\hat{h}) > \varepsilon) \leq \delta$$

or

$$p(\varepsilon_D(\hat{h}) \leq \varepsilon) \geq 1 - \delta$$

Finite Hypothesis Case

- If the hypothesis space H is finite, denote the empirical risk minimization solution as

$$\hat{h} = \arg\min_{h \in H} \hat{\varepsilon}_S(h)$$

- **Theorem:** Assume that H is a Finite Hypothesis class. For any $\varepsilon, \delta \in (0,1)$, the sample complexity is given by

$$N(\varepsilon, \delta) = \frac{\log(|H|/\delta)}{\varepsilon}.$$

- i.e. for any $N \geq N(\varepsilon, \delta)$ sample data, the empirical risk minimization solution $\hat{h} \in H$ satisfies

$$p(\varepsilon_D(\hat{h}) > \varepsilon) \leq \delta.$$

- In other words, $\hat{h} \in H$ the following inequality satisfies with at least probability $1 - \delta$

$$\varepsilon_D(\hat{h}) \leq \varepsilon.$$

Outline of Proof

- Step 1: Prove the following inequality

$$p(\{D \mid \varepsilon_D(\hat{h}) > \varepsilon\} \leq |H| e^{-\varepsilon N}$$

- Step 2: Define $\delta = |H| e^{-\varepsilon N}$ or $N(\varepsilon, \delta) = \dfrac{\log(|H|/\delta)}{\varepsilon}$.

- Step 3: for any $N \geq N(\varepsilon, \delta)$ sample data, the empirical risk minimization solution $\hat{h} \in H$ satisfies

$$p(\varepsilon_D(\hat{h}) > \varepsilon) \leq \delta.$$

Proof

- Step 1: Prove the following inequality $p(\{D \mid \varepsilon_D(\hat{h}) > \varepsilon\} \leq |H| e^{-\varepsilon N}$
- For any $h(x)$ with $\varepsilon_D(h) > \varepsilon$, then

$$p\{x \in D \mid h(x) = c(x)\} = 1 - p\{x \in D \mid h(x) \neq c(x)\} \leq 1 - \varepsilon$$

$$p\left(\bigcap_{x \in D} \{h_i(x) = c(x)\}\right) = \left(1 - p\{h(x) \neq c(x)\}\right)^N \leq (1 - \varepsilon)^N$$

- Accoding to the assumption, the Hypothesis is PAC learnable, there exists $h(x)$ such that

$$\varepsilon(h) = 0$$

- But we don't know which one

$$p(\exists h \in H \mid \varepsilon(h) > \varepsilon, \hat{\varepsilon}(h) = 0) \leq \sum_i p(h_i \in H \mid \varepsilon(h_i) > \varepsilon, \hat{\varepsilon}(h_i) = 0)$$

$$= |H|(1 - \varepsilon)^N \leq |H| e^{-N\varepsilon} \equiv \delta$$

Implications

- Suppose we want this probability to be at most δ

$$p(\varepsilon_D(h) > \varepsilon) \leq \delta.$$

- How many training examples suffice ?

$$N(\varepsilon, \delta) = \frac{1}{\varepsilon}\left(\log(|H|) + \log(1/\delta)\right).$$

- If *errortrain(h) = 0,* then with probability at least 1 - δ

$$\varepsilon(\hat{h}) \leq \frac{1}{N}\left(\log(|H|) + \log(1/\delta)\right).$$

- **Theorem:** Conjunctions of Boolean literals is PAC learnable

Simple Example

If $|H| = 973$, and $m(\varepsilon,\delta) = \frac{1}{\varepsilon}\left(\log(|H|) + \log(1/\delta)\right).$

If want to assure that with probability 95%($\delta = 0.05$), we require the hypotheses with $\text{error}_{D(h)} \leq 0.1$ ($\varepsilon = 0.1$), then it is sufficient to have N examples, where

$$
\begin{aligned}
N \geq m(\varepsilon,\delta) &= \frac{1}{\varepsilon}\left(\log(|H|) + \log(1/\delta)\right) \\
&= \frac{1}{0.1}\left(\log(973) + \log(1/0.05)\right) \\
&= 10\left(\log(973) + \log(20)\right) \approx 98.8
\end{aligned}
$$

Ex 2: Learning conjunctions of boolean literals

- Let H be the hypothesis space defined by conjunctions of literals based on n Boolean attributes possibly with negation

- **Question:** How many examples are sufficient to assure with probability of at least $(1 - \delta)$ that every h in H satisfies error $error_{D(h)} \leq \epsilon$?

- **Answer:** $|H| = 3^n$, and using our theorem it follows that

$$N(\varepsilon,\delta) = \frac{1}{\varepsilon}\left(n\log 3 + \log(1/\delta)\right).$$

Agnostic Learning

- **Agnostic learning setting:** don't assume $c \in H$

- What do we want then?
  » The hypothesis *h* that makes fewest errors on training data

- What is sample complexity in this case ?

$$N(\varepsilon,\delta) = \frac{1}{2\varepsilon^2}\left(\log(|H|) + \log(1/\delta)\right).$$

derived from Hoeffding bounds:

$$p(\varepsilon_D(h) > \hat{\varepsilon}_S(h) + \varepsilon) \le e^{-2N\varepsilon^2}.$$

Empirical Risk Minimization Paradigm

- For an input sample S, find some $h \in H$ that fits S "well"
- For a new point x, predict a label according to its membership in h

$$\hat{h} = \arg\min_{h \in H} \hat{\varepsilon}_S(h)$$

- Example:
  - Consider linear classification, and let $h_\theta(x) = 1(\theta^T x \geq 0)$

$$H = \left\{ h_\theta : \; h_\theta(x) = 1(\theta^T x \geq 0), \quad \theta \in R^{n+1} \right\}$$

  - Then

$$\hat{\theta} = \arg\min_\theta \hat{\varepsilon}_S(h_\theta)$$

- In learning theory, it will be useful to abstract away from the specific parameterization of hypotheses and from issues such as whether we are using a linear classifier or an ANN

The Case of Finite H

- H = $\{h_1, \ldots, h_k\}$ are k hypotheses. How to derive the generalization error of $\hat{h} \in H$
- First, we will show that $\hat{\varepsilon}(\hat{h})$ is a reliable estimate of $\varepsilon(h)$ for all h.
- Second, we will show that this implies an upper-bound on the generalization error of $\hat{h} \in H$

The outcome of a binary classifier can be viewed as a Bernoulli random variable Z:

$$Z = 1\{\, h(x) \neq c(x) \,\}$$

- For each sample: $Z_i = 1\{\hat{h}(x_i) \neq c(x_i)\}$
- The training error:

$$\hat{\varepsilon}(\hat{h}) = \frac{1}{N}\sum_{i=1}^{N} Z_i$$

Hoeffding Inequality

- **Lemma.** (Hoeffding Inequality) Let $Z_1, Z_2, \ldots, Z_N$ be N independent and identically distributed (iid) random variables drawn from a Bernoulli (φ) distribution, i.e., $P(Z_i = 1) = \varphi$ and $P(Z_i = 0) = 1 - \varphi$.

Let $\hat{\phi} = \frac{1}{N}\sum_i Z_i$ be the mean of these random variables. For any given $\varepsilon > 0$, we have

$$p(\left|\phi - \hat{\phi}\right| > \varepsilon) \leq 2e^{-2\varepsilon^2 N}$$

  - This lemma (which in learning theory is also called the **Chernoff bound**) says that if we take $\hat{\phi}$ — the average of N Bernoulli (ϕ) random variables — to be our estimate of φ, then the probability of our being far from the true value is small, so long as N is large.

Misclassification Probability

- **The outcome of a binary classifier** can be viewed as a Bernoulli random variable Z:

$$Z = 1\{\, h(\mathrm{x}) \neq \mathrm{c}(\mathrm{x}) \,\}$$

- For each sample: $Z_i = 1\{\, \hat{h}(x_i) \neq c(x_i) \,\}$
- The training error:

$$\hat{\varepsilon}(\hat{h}) = \frac{1}{N}\sum_{i=1}^{N} Z_i$$

- Hoeffding inequality $p(|\, \varepsilon(\hat{h}) - \hat{\varepsilon}(\hat{h}) \,| > \varepsilon) \leq 2e^{-2\varepsilon^2 N}$

This shows that, for our particular h_i, training error will be close to generalization error with high probability, assuming m is large.

Uniform Convergence

- But we don't just want to guarantee that $\hat{\varepsilon}(\hat{h})$ will be close to $\varepsilon(\hat{h})$ (with high probability) for just only one particular h_i.
- We want to prove that this will be true for simultaneously for all $h_i \in$ H
- For k hypothesis, denote $A_i = \{\varepsilon(h_i) - \hat{\varepsilon}(h_i) |> \varepsilon\}$

$$p(\exists h \in H, |\varepsilon(h_i) - \hat{\varepsilon}(h_i)| > \varepsilon) = p(A_1 \cup \cdots \cup A_k)$$

$$\leq \sum_{i=1}^{k} p(A_i) \leq \sum_{i=1}^{k} 2e^{-2\varepsilon^2 N} = 2ke^{-2\varepsilon^2 N}$$

- In other words

$$p(\neg\exists h \in H, |\varepsilon(h_i) - \hat{\varepsilon}(h_i)| > \varepsilon) = p(\forall h \in H, |\varepsilon(h_i) - \hat{\varepsilon}(h_i)| \leq \varepsilon)$$

$$\leq 1 - 2ke^{-2\varepsilon^2 N}$$

Sample Complexity

How many training examples N we need in order to make a guarantee ?

$$p(\exists h \in H, |\varepsilon(h_i) - \hat{\varepsilon}(h_i)| > \varepsilon) \leq 2ke^{-2\varepsilon^2 N} = \delta$$

We find that if $N \geq \frac{1}{2\varepsilon^2} \log\left(\frac{2k}{\delta}\right)$, then with probability at least $1-\delta$, we have that

$$\left|\varepsilon(h_i) - \hat{\varepsilon}(h_i)\right| \leq \varepsilon$$

for all $h_i \in$ H

- The key property of the bound above is that the number of training examples needed to make this guarantee is only **logarithmic in *k***, the number of hypotheses in H.

Generalization Error Bound

- We can also hold N and δ fixed and solve for ε in the previous equation, and show that with probability $1-\delta$,

$$\left|\hat{\varepsilon}(h)-\varepsilon(h)\right| \le \sqrt{\frac{1}{2N}\log\left(\frac{2k}{\delta}\right)}$$

for all $h_i \in H$.

Denote $h^* = \arg\min_{h\in H} \varepsilon(h)$ to be the best hypothesis in H. And $\hat{h} = \arg\min_{h\in H} \hat{\varepsilon}(h)$

$$\begin{aligned}\varepsilon(\hat{h}) &\le \hat{\varepsilon}(\hat{h}) + \varepsilon \le \hat{\varepsilon}(h^*) + \varepsilon \\ &\le \varepsilon(h^*) + 2\varepsilon\end{aligned}$$

If uniform convergence occurs, then the generalization error of $\varepsilon(\hat{h})$ is at most 2ε worse than the best possible hypothesis in H !

Summary

- **Theorem.** Let $|H| = k$, and let any N, δ be fixed. Then with probability at least $1-\delta$ we have that

$$\varepsilon(\hat{h}) \le \min_{h \in H} \varepsilon(h) + 2\sqrt{\frac{1}{2N}\log\left(\frac{2k}{\delta}\right)}$$

Corollary. Let |H| = k, and let any δ, ε be fixed. Then for

$$\varepsilon(\hat{h}) \le \min_{h \in H} \varepsilon(h) + 2\varepsilon$$

to hold with probability at least $1-\delta$, it suffices that

$$N \ge \frac{1}{2\varepsilon^2}\log\left(\frac{2k}{\delta}\right) = O\left(\frac{1}{2\varepsilon^2}\log\left(\frac{k}{\delta}\right)\right)$$

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